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United States Patent	12406922
Kind Code	B2
Date of Patent	September 02, 2025
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Semiconductor structure and manufacturing method thereof

Abstract

A semiconductor structure including a substrate and a pad structure is provided. The pad structure is located on the substrate. The pad structure includes material pairs and pads. The material pairs are stacked on the substrate to form a stair step structure. Each of the material pairs includes a conductive layer and a dielectric layer located on the conductive layer. Each of the pads includes a conductive pillar and a pad layer. The conductive pillar is embedded in the material pair and is connected to the conductive layer of the material pair. The pad layer is located on the conductive pillar.

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Appl. No.:	17/900587
Filed:	August 31, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240071906 A1	Feb. 29, 2024

Publication Classification

Int. Cl.:	H01L23/522 (20060101); H01L21/768 (20060101)
U.S. Cl.:	
CPC	H01L23/5226 (20130101); H01L21/768 (20130101);

Field of Classification Search

CPC: H01L (23/5226); H01L (21/768)

USPC: 257/773

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Background/Summary

BACKGROUND

Technical Field

(1) The invention relates to a semiconductor structure and a manufacturing method thereof, and particularly relates to a semiconductor structure having a pad structure and a manufacturing method thereof.

Description of Related Art

(2) As the integrity of the memory device increases, in order to achieve high density and high performance, it is more and more common to replace the two-dimensional (2D) memory devices with the three-dimensional (3D) memory devices. One of the 3D memory devices is the vertical memory device. Although the vertical memory device can increase the memory capacity per unit area, the interconnection difficulty in the vertical memory device also increases.

(3) In general, conductive layers with a stair step structure often serves as pads in the 3D memory device, and the pad and the contact thereon often serve as the interconnection structure to connect the component in each layer to other component. However, during the etching process for forming the contact opening, since the pads at different positions in the stair step structure have different distances from the top surface of the dielectric layer on the pads, the topmost pad of the stair step structure may be over-etched, and thus the contact opening penetrates the topmost pad and extend to the conductive layer below the topmost pad. In this way, the electrical failure of the device will

be caused due to the undesired bridging between the subsequently formed contact and other pad or conductive layer. Therefore, how to prevent the electrical failure caused by over-etching during the process of manufacturing the contact opening is an important issue at present.

SUMMARY

(4) The invention provides a semiconductor structure and a manufacturing method thereof, which can prevent the electrical failure caused by over-etching during the process of manufacturing the contact opening.

(5) The invention provides a semiconductor structure, which includes a substrate and a pad structure. The pad structure is located on the substrate. The pad structure includes material pairs and pads. The material pairs are stacked on the substrate to form a stair step structure. Each of the material pairs includes a conductive layer and a dielectric layer located on the conductive layer. Each of the pads includes a conductive pillar and a pad layer. The conductive pillar is embedded in the material pair and is connected to the conductive layer of the material pair. The pad layer is located on the conductive pillar.

(6) According to an embodiment of the invention, in the semiconductor structure, the thickness of the pad may be greater than the thickness of the conductive layer.

(7) According to an embodiment of the invention, in the semiconductor structure, the thickness of the pad may be 2 to 4 times the thickness of the conductive layer.

(8) According to an embodiment of the invention, in the semiconductor structure, the thickness of the pad may be greater than the thickness of the material pair.

(9) According to an embodiment of the invention, in the semiconductor structure, the conductive layer, the conductive pillar, and the pad layer may be integrally formed.

(10) According to an embodiment of the invention, in the semiconductor structure, the bottom surface of the conductive pillar and the bottom surface of the conductive layer may be coplanar.

(11) According to an embodiment of the invention, in the semiconductor structure, the pad layer may be located on the top surface of the dielectric layer on two sides of the conductive pillar.

(12) According to an embodiment of the invention, in the semiconductor structure, the width of the pad layer may be greater than the width of the conductive pillar.

(13) According to an embodiment of the invention, in the semiconductor structure, the thickness of the pad layer may be greater than or equal to half of the width of the conductive pillar.

(14) According to an embodiment of the invention, in the semiconductor structure, the top-view pattern of the conductive pillars may include a polygon, a circle, or an ellipse.

(15) According to an embodiment of the invention, in the semiconductor structure, the size of the top-view pattern of the pad layer may be greater than the size of the top-view pattern of the conductive pillar.

(16) According to an embodiment of the invention, in the semiconductor structure, there may be a gap between the edge of the top-view pattern of the pad layer and the edge of the top-view pattern of the conductive pillar, and the gap may surround the top-view pattern of the conductive pillar.

(17) According to an embodiment of the invention, in the semiconductor structure, two opposite side edges of the top-view pattern of the pad layer and two opposite side edges of the top-view pattern of the conductive pillar may be flush with each other.

(18) According to an embodiment of the invention, the semiconductor structure may further include contacts. The contact may be electrically connected to the pad.

(19) According to an embodiment of the invention, in the semiconductor structure, the width of the pad layer may be greater than the width of the bottom of the contact.

(20) According to an embodiment of the invention, in the semiconductor structure, the contacts in different regions may have different landing depths.

(21) According to an embodiment of the invention, the semiconductor structure may further include a protection layer. The protection layer is located between the stair step structure and the substrate.

(22) According to an embodiment of the invention, in the semiconductor structure, the edge of the

pad layer may exceed the edge of the conductive layer.

(23) According to an embodiment of the invention, in the semiconductor structure, the horizontal spacing between two adjacent pad layers may be substantially equal to the thickness of the pad layer.

(24) The invention provides a manufacturing method of a semiconductor structure, which includes the following steps. A substrate is provided. A pad structure is formed on the substrate. The pad structure includes material pairs and pads. The material pairs are stacked on the substrate to form a first stair step structure. Each of the material pairs includes a conductive layer and a first dielectric layer located on the conductive layer. Each of the pads includes a conductive pillar and a pad layer. The conductive pillar is embedded in the material pair and is connected to the conductive layer of the material pair. The pad layer is located on the conductive pillar.

(25) Based on the above description, in the semiconductor structure and the manufacturing method thereof according to the invention, each of the pads includes the conductive pillar and the pad layer, the conductive pillar is embedded in the material pair and is connected to the conductive layer of the material pair, and the pad layer is located on the conductive pillar. Therefore, compared with the conventional pad, the pad of the invention can have greater thickness, thereby preventing the electrical failure caused by over-etching during the process of manufacturing the contact opening. In addition, since the thicker pad is used as the etching stop layer during formation of the contact opening, the process window of the contact opening can be improved and the process yield can be increased.

(26) In order to make the aforementioned and other objects, features and advantages of the invention comprehensible, several exemplary embodiments accompanied with drawings are described in detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are included to provide a further understanding of the invention, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

(2) FIG. 1A to FIG. 1Q are cross-sectional views illustrating a manufacturing process of a semiconductor structure according to some embodiments of the invention.

(3) FIG. 1R is cross-sectional view illustrating a semiconductor structure according to other embodiments of the invention.

(4) FIG. 2A to FIG. 2C are top views illustrating some stages of a manufacturing process of a semiconductor structure according to some embodiments of the invention.

(5) FIG. 3A to FIG. 3C are top views illustrating some stages of a manufacturing process of a semiconductor structure according to other embodiments of the invention.

(6) FIG. 4A to FIG. 4C are top views illustrating some stages of a manufacturing process of a semiconductor structure according to other embodiments of the invention.

DESCRIPTION OF THE EMBODIMENTS

(7) The embodiments are described in detail below with reference to the accompanying drawings, but the embodiments are not intended to limit the scope of the invention. For the sake of easy understanding, the same components in the following description will be denoted by the same reference symbols. In addition, the drawings are for illustrative purposes only and are not drawn to the original dimensions. Furthermore, the features in the top view and the features in the cross-sectional view are not drawn to the same scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(8) FIG. 1A to FIG. 1Q are cross-sectional views illustrating a manufacturing process of a semiconductor structure according to some embodiments of the invention. FIG. 1R is cross-sectional view illustrating a semiconductor structure according to other embodiments of the invention. FIG. 2A to FIG. 2C are top views illustrating some stages of a manufacturing process of a semiconductor structure according to some embodiments of the invention. FIG. 1A to FIG. 1R are cross-sectional views taken along section line I-I' in FIG. 2A to FIG. 2C. FIG. 3A to FIG. 3C are top views illustrating some stages of a manufacturing process of a semiconductor structure according to other embodiments of the invention. FIG. 4A to FIG. 4C are top views illustrating some stages of a manufacturing process of a semiconductor structure according to other embodiments of the invention. In the top view of the present embodiment, some components in the cross-sectional view are omitted to clearly illustrate the configuration relationship between the components in the top view.

(9) Referring to FIG. 1A, a substrate **100** is provided. In some embodiments, the substrate **100** may be a semiconductor substrate (e.g., silicon substrate), a compound semiconductor substrate (e.g., gallium arsenide substrate), or a semiconductor-on-insulator (SOI) substrate, but the invention is not limited thereto. In addition, although not shown in FIG. 1A, the substrate **100** may have required components thereon, such as semiconductor devices (e.g., active devices and/or passive devices), interconnect structures and/or dielectric layers, and the description thereof is omitted here.

(10) A protection layer **102** may be formed on the substrate **100**. In some embodiments, the material of the protection layer **102** is, for example, silicon oxide. In some embodiments, the method of forming the protection layer **102** is, for example, a chemical vapor deposition (CVD) method.

(11) A stack structure **104** may be formed on substrate **100**. In some embodiments, the stack structure **104** may be formed on the protection layer **102**. The stack structure **104** includes sacrificial layers **106** (e.g., sacrificial layers **106a-106j**) and dielectric layers **108** (e.g., dielectric layers **108a-108j**) alternately stacked. In some embodiments, the material of the sacrificial layer **106** is, for example, silicon nitride. In some embodiments, the method of forming the sacrificial layer **106** is, for example, a CVD method. In some embodiments, the material of the dielectric layer **108** is, for example, silicon oxide. In some embodiments, the method of forming the dielectric layer **108** is, for example, a CVD method.

(12) A stop layer **110** may be formed on the stack structure **104**. In some embodiments, the material of the stop layer **110** is, for example, polysilicon. In some embodiments, the method of forming the stop layer **110** is, for example, a CVD method.

(13) Referring to FIG. 1B, the stop layer **110** may be patterned to expose the stack structure **104**. For example, the stop layer **110** may expose the dielectric layer **108j** of the stack structure **104**. In some embodiments, the stop layer **110** may be patterned by a lithography process and an etching process.

(14) Referring to FIG. 1C, a patterned photoresist layer **112** may be formed on the stack structure **104** and the stop layer **110**. In some embodiments, the patterned photoresist layer **112** may be formed by a lithography process.

(15) Referring to FIG. 1D, a portion of the dielectric layer **108j** and a portion of the sacrificial layer **106j** are removed by using the patterned photoresist layer **112** as a mask to form openings **114** (e.g., openings **114a-114j**) in the dielectric layer **108j** and the sacrificial layer **106j**. In some embodiments, the method of removing a portion of the dielectric layer **108j** and a portion of the sacrificial layer **106j** is, for example, a dry etching method.

(16) The patterned photoresist layer **112** may be removed. In some embodiments, the method of removing the patterned photoresist layer **112** is, for example, a dry stripping method or a wet stripping method.

(17) Referring to FIG. 1E, a patterned photoresist layer **116** may be formed on the stack structure

104 and the stop layer **110**. The patterned photoresist layer **116** may expose the opening **114a** and a portion of the stack structure **104**. In some embodiments, the patterned photoresist layer **116** may be formed by a lithography process.

(18) Referring to FIG. **1F**, an etching process is performed by using the patterned photoresist layer **116** as a mask to remove a portion of the dielectric layer **108j** and a portion of the sacrificial layer **106j** exposed by the patterned photoresist layer **116** and a portion of the dielectric layer **108i** and a portion of the sacrificial layer **106i** located below the opening **114a**, so as to transfer the pattern of the opening **114a** into the dielectric layer **108i** and the sacrificial layer **106i**. Therefore, the opening **114a** transferred into the dielectric layer **108i** and the sacrificial layer **106i** may expose the dielectric layer **108h**. In addition, in this step, the patterned photoresist layer **116** may be etched, so that the thickness of the patterned photoresist layer **116** may be reduced. In some embodiments, the etching process may be a dry etching process such as a reactive ion etching (RIE) process.

(19) Referring to FIG. **1G**, a trimming process may be performed on the patterned photoresist layer **116** to expose the opening **114b**. The trimming process refers to a process of pulling back the patterned photoresist layer **116** by a distance **D1**. In this way, as shown in FIG. **1G**, the patterned photoresist layer **116** may expose the opening **114a** and the opening **114b**. In some embodiments, the thickness of the patterned photoresist layer **116** is reduced when the patterned photoresist layer **116** is trimmed and pulled back. In some embodiments, the trimming process may be performed by a dry etching process.

(20) Referring to FIG. **1H**, an etching process is performed by using the patterned photoresist layer **116** as a mask to remove a portion of the dielectric layer **108j**, a portion of the sacrificial layer **106j**, a portion of the dielectric layer **108i**, a portion of the sacrificial layer **106i**, a portion of the dielectric layer **108h**, and a portion of the sacrificial layer **106h**, so as to transfer the pattern of the opening **114a** into the dielectric layer **108h** and the sacrificial layer **106h** and to transfer the pattern of the opening **114b** into the dielectric layer **108i** and the sacrificial layer **106i**. Therefore, the opening **114a** transferred into the dielectric layer **108h** and the sacrificial layer **106h** may expose the dielectric layer **108g**, and the opening **114b** transferred into the dielectric layer **108i** and the sacrificial layer **106i** may expose the dielectric layer **108h**. In addition, in this step, the patterned photoresist layer **116** may be etched, so that the thickness of the patterned photoresist layer **116** may be reduced. In some embodiments, the etching process may be a dry etching process such as a reactive ion etching (RIE) process.

(21) Referring to FIG. **1I**, a trimming process may be performed on the patterned photoresist layer **116** to expose the opening **114c**. The trimming process may pull back the patterned photoresist layer **116** by a distance **D2**. In this way, as shown in FIG. **1I**, the patterned photoresist layer **116** may expose the opening **114a**, the opening **114b**, and the opening **114c**. In some embodiments, the thickness of the patterned photoresist layer **116** is reduced when the patterned photoresist layer **116** is trimmed and pulled back. In some embodiments, the trimming process may be performed by a dry etching process.

(22) Referring to FIG. **1J**, the etching process and the trimming process may be repeated until the stair step structure **104a** shown in FIG. **1J** is formed. By the above method, the stack structure **104** may be patterned to form a stair step structure **104a** and to form the opening **114** in each of stair steps of the stair step structure **104a**. In some embodiments, the opening **114** may pass through one dielectric layer **108** and one sacrificial layer **106**.

(23) Furthermore, after forming the stair step structure **104a**, the patterned photoresist layer may be removed. In some embodiments, the method of removing the patterned photoresist layer **116** is, for example, a dry stripping method or a wet stripping method.

(24) In some embodiments, the top-view pattern of the opening **114** may be a polygon, a circle, or an ellipse. Referring to FIG. **1J** and FIG. **2A**, the top-view pattern of the opening **114** may be a square shape, but the invention is not limited thereto. In other embodiments, as shown in FIG. **3A** and FIG. **4A**, the top-view pattern of the opening **114** may be a rectangle shape (FIG. **3A**) or a bar

shape (FIG. 4A).

(25) Referring to FIG. 1K, a sacrificial layer **118** may be conformally formed on the upper surfaces **S1** and the side surfaces **S2** of the stair step structure **104a** and in the openings **114**. In some embodiments, the sacrificial layer **118** may be further conformally formed on stop layer **110**. In some embodiments, the thickness **T1** of the sacrificial layer **118** may be greater than or equal to half of the width **W1** of the opening **114**. In some embodiments, the material of the sacrificial layer **118** is, for example, silicon nitride. In some embodiments, the method of forming the sacrificial layer **118** is, for example, a CVD method.

(26) Referring to FIG. 1L, a surface hardening treatment may be performed on the sacrificial layer **118** to form sacrificial layers **120**. The sacrificial layers **120** cover the sacrificial layer **118** located in the openings **114** and expose the sacrificial layer **118** located on the side surfaces **S2** of the stair step structure **104a**. In some embodiments, the sacrificial layer may cover the stop layer **110**. In some embodiments, the surface hardening treatment may be an argon (Ar) plasma treatment.

(27) Referring to FIG. 1M, the sacrificial layer **118** located on the side surfaces **S2** of the stair step structure **104a** may be removed. In some embodiments, in the step of removing the sacrificial layer **118** located on the side surfaces **S2** of the stair step structure **104a**, a portion of the sacrificial layer **120** may be removed, but the invention is not limited thereto. In some embodiments, the method of removing the sacrificial layer **118** located on the side surfaces **S2** of the stair step structure **104a** is, for example, a wet etching method. For example, the sacrificial layer **118** located on the side surfaces **S2** of the stair step structure **104a** may be removed by hot phosphoric acid.

(28) Referring to FIG. 1N, a dielectric layer **122** may be formed on the stair step structure **104a** and the sacrificial layers **120**. The dielectric layer **122** may be a single-layer structure or a multilayer structure. In some embodiments, the material of the dielectric layer **122** is, for example, silicon oxide. In some embodiments, the method for forming the dielectric layer **122** is, for example, a CVD method. In some embodiments, during the formation of the dielectric layer **122**, the stop layer **110** and the sacrificial layer **120** located on the stop layer **110** may be removed.

(29) In some embodiments, the top-view pattern of the sacrificial layer **118** located in the opening **114** may be a polygon, a circle, or an ellipse. Referring to FIG. 1N and FIG. 2B, the top-view pattern of the sacrificial layer **118** located in the opening **114** may be a square shape, but the invention is not limited thereto. In other embodiments, as shown in FIG. 3B and FIG. 4B, the top-view pattern of the sacrificial layer **118** located in the opening **114** may be a rectangle shape (FIG. 3B) or a bar shape (FIG. 4B). In addition, referring to FIG. 2B, FIG. 3B, and FIG. 4B, the top-view pattern of the sacrificial layer **120** may be a bar shape.

(30) Referring to FIG. 1O and FIG. 2C, a replacement process may be performed to replace the sacrificial layer **106**, the sacrificial layer **118**, and the sacrificial layer **120** with a conductive layer **124**, a conductive pillar **126**, and a pad layer **128**, respectively. For example, the replacement process may include the following steps. First, slits **SLT** (FIG. 2C) may be formed in the dielectric layer **122** and the stair step structure **104a**. Although the cross-sectional view of FIG. 1O does not show the slits **SLT**, it can be learned from FIG. 2C that the extension direction of the slit **SLT** may be parallel to the direction of the section line I-I'. The slits **SLT** may extend to the bottom surface of the stair step structure **104a**, so as to expose a cross-section of the sacrificial layer **106** and a cross-section of the sacrificial layer **120**. Then, the sacrificial layer **106**, the sacrificial layer **118**, and the sacrificial layer **120** may be removed to form gaps (not shown). In some embodiments, the method of removing the sacrificial layer **106**, the sacrificial layer **118**, and the sacrificial layer **120** is, for example, a wet etching method. For example, an etchant may be applied in the slits **SLT** to remove the sacrificial layer **106**, the sacrificial layer **118**, and the sacrificial layer **120**. In some embodiments, the etchant is, for example, hot phosphoric acid. Next, a deposition process is performed to form a conductive material layer (not shown) in the gaps and the slits **SLT**. The conductive material layer may have a single-layer structure or a multilayer structure. In some embodiments, the material of the conductive material layer is, for example, tungsten, titanium,

titanium nitride, or a combination thereof, but the invention is not limited thereto. Then, the conductive material layer in the slits SLT may be removed by an etch-back process to form the conductive layer **124**, the conductive pillar **126**, and the pad layer **128**.

(31) In this way, as shown in FIG. **10**, after the replacement process is performed, the sacrificial layer **106** is replaced with the conductive layer **124**, the sacrificial layer **118** is replaced with the conductive pillar **126**, and the sacrificial layer **120** is replaced with the pad layer **128**. In some embodiments, the conductive layer **124**, the conductive pillar **126**, and the pad layer **128** may be formed simultaneously by the same process. In some embodiments, the conductive layer **124**, the conductive pillar **126**, and the pad layer **128** may be integrally formed. In some embodiments, the materials of the conductive layer **124**, the conductive pillar **126**, and the pad layer **128** may be the same. Each of the conductive layer **124**, the conductive pillar **126**, and the pad layer **128** may be a single-layer structure or a multilayer structure. In some embodiments, the materials of the conductive layer **124**, the conductive pillar **126**, and the pad layer **128** are, for example, tungsten, titanium, titanium nitride, or a combination thereof, but the invention is not limited thereto.

(32) By the above method, a pad structure **130** may be formed on the substrate **100**. The pad structure **130** includes material pairs **132** and pads **134** (e.g., pads **134a-134j**). The material pairs **132** are stacked on the substrate **100** to form a stair step structure **104b**. Each of the material pair **132** includes a conductive layer **124** and a dielectric layer **108** located on the conductive layer **124**. Each of the pads **134** includes a conductive pillar **126** and a pad layer **128**. The conductive pillar **126** is embedded in the material pair **132** and is connected to the conductive layer **124** of the material pair **132**. The pad layer **128** is located on the conductive pillar **126**. The pad layer **128** may be located on the top surface S3 of the dielectric layer **108** of the material pair **132**. In some embodiments, the protection layer **102** may be formed between the stair step structure **104b** and the substrate **100**.

(33) Referring to FIG. **1P**, contact openings **136** (e.g., contact openings **136a-136j**) may be formed in the dielectric layer **122**. The contact openings **136a** to **136j** expose the pads **134a** to **134j**, respectively. In some embodiments, the contact opening **136** may be formed by patterning the dielectric layer **122** by a lithography process and an etching process. In some embodiments, the pads **134a-134j** may be used as etching stop layers during formation of the contact openings **136a-136j**. Compared with the distance between the top surface S4 of the pad **134a** and the top surface S5 of the dielectric layer **122**, the distance between the top surface S6 of the pad **134j** and the top surface S5 of the dielectric layer **122** is shorter. In this way, during the process of forming the contact openings **136**, the contact opening **136j** firstly touches the top surface S6 of the top pad **134j**, so that the etching amount of the top pad **134j** is greater than the etching amounts of other pads **134a-134i**. Since the pad **134** of the above embodiments can have greater thickness, the electrical failure caused by over-etching during the process of manufacturing the contact opening **136** can be prevented. In addition, since the thicker pad **134** is used as the etching stop layer during formation of the contact opening **136**, the process window of the contact opening **136** can be improved and the process yield can be increased.

(34) In some embodiments, other required processes may be performed before the contact openings **136** are formed, and the description thereof is omitted here.

(35) Referring to FIG. **1Q**, contacts **138** (e.g., contacts **138a-138j**) may be formed in the contact openings **136** (e.g., contact openings **136a-136j**). Therefore, the contact **138** electrically connected to the pad **134** may be formed in the dielectric layer **122**. For example, the contacts **138a-138j** may be electrically connected to the pads **134a-134j**, respectively. In some embodiments, the method of forming the contacts **138** may include the following steps. First, a contact material layer (not shown) filling the contact openings **136** may be formed. The contact material layer may be a single-layer structure or a multilayer structure. In some embodiments, the material of the contact material layer is, for example, tungsten, titanium, titanium nitride, or a combination thereof, but the invention is not limited thereto. In some embodiments, the method of forming the contact material

layer is, for example, a physical vapor deposition (PVD) method or a CVD method. Then, the contact material layer outside the contact openings **136** may be removed to form the contacts **138**. In some embodiments, the method of removing the contact material layer outside the contact openings **136** is, for example, a chemical mechanical polishing (CMP) method. In some embodiments, the top surface **S7** of the contact **138** and the top surface **S5** of the dielectric layer **122** may be coplanar. In some embodiments, the material of the contact **138** and the material of the pad **134** may be the same. In other embodiments, the material of the contact **138** and the material of the pad **134** may be different. The contact **138** may be a single-layer structure or a multilayer structure. In some embodiments, the material of the contact **138** is, for example, tungsten, titanium, titanium nitride, or a combination thereof, but the invention is not limited thereto.

(36) Hereinafter, the semiconductor structure **10** of the present embodiment is described with reference to FIG. **1Q**. In addition, although the method for forming the semiconductor structure **10** is described by taking the above method as an example, the invention is not limited thereto.

(37) Referring to FIG. **1Q**, a semiconductor structure **10** includes the substrate **100** and the pad structure **130**. In some embodiments, the semiconductor structure **10** may be a 3D memory structure, but the invention is not limited thereto.

(38) The pad structure **130** is located on the substrate **100**. The pad structure **130** includes the material pairs **132** and the pads **134**. The material pairs **132** are stacked on the substrate to form the stair step structure **104b**. Each of the material pairs **132** includes the conductive layer **124** and the dielectric layer **108** located on the conductive layer **124**. Each of the pads **134** includes the conductive pillar **126** and the pad layer **128**. The conductive pillar is embedded in the material pair **132** and is connected to the conductive layer **124** of the material pair **132**. The conductive pillar **126** may pass through the dielectric layer **108**. The pad layer **128** is located on the conductive pillar **126**. The pad layer **128** may be located and on the top surface **S3** of the dielectric layer **108** of the material pair **132**. In some embodiments, the pad layer **128** may be located on the top surface **S3** of the dielectric layer **108** on two sides of the conductive pillar **126**. In some embodiments, the cross-sectional shape of the pad **134** may be a T-shape. In some embodiments, the conductive layer **124**, the conductive pillar **126**, and the pad layer **128** may be integrally formed. In some embodiments, the bottom surface **S8** of the conductive pillar **126** and the bottom surface **S9** of the conductive layer **124** may be coplanar.

(39) In some embodiments, the thickness **T2** of the pad **134** may be greater than the thickness **T3** of the conductive layer **124**. In some embodiments, the thickness **T2** of the pad may be 2 to 4 times the thickness **T3** of the conductive layer **124**. In some embodiments, the thickness **T2** of the pad **134** may be greater than the thickness **T4** of the material pair **132**. In some embodiments, the width **W2** of the pad layer **128** may be greater than the width **W3** of the conductive pillar **126**. In some embodiments, the thickness **T5** of the pad layer **128** may be greater than or equal to half of the width **W3** of the conductive pillar **126**.

(40) In some embodiments, the horizontal spacing **SS** between two adjacent pad layers **128** may be substantially equal to the thickness **T5** of the pad layer **128**. In some embodiments, the horizontal spacing **SS** may range from 90% to 110% of the thickness **T5** of the pad layer **128**.

(41) In some embodiments, as shown in FIG. **1Q**, the edge **E1** of the pad layer **128** and the edge **E2** of the conductive layer **124** may be flush with each other, but the invention is not limited thereto. In other embodiments, as shown in FIG. **1R**, the edge **E1** of the pad layer **128** may exceed the edge **E2** of the conductive layer **124**.

(42) In some embodiments, the top-view pattern of the conductive pillar **126** may be a polygon, a circle, or an ellipse. In some embodiments, as shown in FIG. **2C**, FIG. **3C**, and FIG. **4C**, the top-view pattern of the conductive pillar **126** may be a square shape (FIG. **2C**) or a rectangle shape (FIG. **3C** and FIG. **4C**), but the invention is not limited thereto. In some embodiments, as shown in FIG. **2C**, FIG. **3C**, and FIG. **4C**, the size of the top-view pattern of the pad layer **128** may be greater than the size of the top-view pattern of the conductive pillar **126**. In some embodiments, as shown

in FIG. 2C, there may be a gap G between the edge of the top-view pattern of the pad layer **128** and the edge of the top-view pattern of the conductive pillar **126**, and the gap G may surround the top-view pattern of the conductive pillar **126**. In some embodiments, as shown in FIG. 4C, two opposite side edges **S10** of the top-view pattern of the pad layer **128** and two opposite side edges **S11** of the top-view pattern of the conductive pillar **126** may be flush with each other.

(43) In some embodiments, the semiconductor structure **10** may further include the protection layer **102**. The protection layer **102** is located between the stair step structure **104b** and the substrate **100**. In some embodiments, the semiconductor structure **10** may further include the dielectric layer **122**. The dielectric layer **122** covers the pad structure **130**. In some embodiments, the semiconductor structure **10** may further include the contacts **138**. The contacts **138** may be located in the dielectric layer **122**. The contact **138** may be electrically connected to the pad **134**. In some embodiments, the width **W2** of the pad layer **128** may be greater than the width **W4** of the bottom of the contact **138**. In some embodiments, the contacts in different regions may have different landing depths.

(44) Moreover, the details (e.g., the material, the forming method, and the effect) of each component in the semiconductor structure **10** have been described in detail in the above embodiments, and the description thereof is not repeated here.

(45) Based on the above embodiments, in the semiconductor structure **10** and the manufacturing method thereof, each of the pads **134** includes the conductive pillar **126** and the pad layer **128**, the conductive pillar **126** is embedded in the material pair **132** and is connected to the conductive layer **124** of the material pair **132**, and the pad layer **128** is located on the conductive pillar **126**.

Therefore, compared with the conventional pad, the pad **134** of the above embodiments can have greater thickness, thereby preventing the electrical failure caused by over-etching during the process of manufacturing the contact opening **136**. In addition, since the thicker pad **134** is used as the etching stop layer during formation of the contact opening **136**, the process window of the contact opening **136** can be improved and the process yield can be increased.

(46) In summary, in the semiconductor structure and the manufacturing method thereof of the aforementioned embodiments, each of the pads includes the conductive pillar and the pad layer, the conductive pillar is embedded in the material pair and is connected to the conductive layer of the material pair, and the pad layer is located on the conductive pillar. Therefore, compared with the conventional pad, the pad of the aforementioned embodiments can have greater thickness, thereby preventing the electrical failure caused by over-etching during the process of manufacturing the contact opening. In addition, since the thicker pad is used as the etching stop layer during formation of the contact opening, the process window of the contact opening can be improved and the process yield can be increased.

(47) Although the invention has been described with reference to the above embodiments, it will be apparent to one of ordinary skill in the art that modifications to the described embodiments may be made without departing from the spirit of the invention. Accordingly, the scope of the invention is defined by the attached claims not by the above detailed descriptions.

Claims

1. A semiconductor structure, comprising: a substrate; and a pad structure located on the substrate and comprising: material pairs stacked on the substrate to form a stair step structure, wherein each of the material pairs comprises a conductive layer and a dielectric layer located on the conductive layer; and pads, wherein each of the pads comprises: a conductive pillar embedded in the material pair and connected to the conductive layer of the material pair; and a pad layer located on the conductive pillar, wherein a thickness of the pad is greater than a thickness of the conductive layer.
2. The semiconductor structure according to claim 1, wherein a thickness of the pad is 2 to 4 times a thickness of the conductive layer.
3. The semiconductor structure according to claim 1, wherein a thickness of the pad is greater than

a thickness of the material pair.

4. The semiconductor structure according to claim 1, wherein the conductive layer, the conductive pillar, and the pad layer are integrally formed.

5. The semiconductor structure according to claim 1, wherein a bottom surface of the conductive pillar and a bottom surface of the conductive layer are coplanar.

6. The semiconductor structure according to claim 1, wherein the pad layer is located on a top surface of the dielectric layer on two sides of the conductive pillar.

7. The semiconductor structure according to claim 1, wherein a width of the pad layer is greater than a width of the conductive pillar.

8. The semiconductor structure according to claim 1, wherein a thickness of the pad layer is greater than or equal to half of a width of the conductive pillar.

9. The semiconductor structure according to claim 1, wherein a top-view pattern of the conductive pillar comprises a polygon, a circle, or an ellipse.

10. The semiconductor structure according to claim 1, wherein a size of a top-view pattern of the pad layer is greater than a size of a top-view pattern of the conductive pillar.

11. The semiconductor structure according to claim 10, wherein there is a gap between an edge of the top-view pattern of the pad layer and an edge of the top-view pattern of the conductive pillar, and the gap surrounds the top-view pattern of the conductive pillar.

12. The semiconductor structure according to claim 1, wherein two opposite side edges of a top-view pattern of the pad layer and two opposite side edges of a top-view pattern of the conductive pillar are flush with each other.

13. The semiconductor structure according to claim 1, further comprising: contacts, wherein the contact is electrically connected to the pad.

14. The semiconductor structure according to claim 13, wherein a width of the pad layer is greater than a width of a bottom of the contact.

15. The semiconductor structure according to claim 13, wherein the contacts in different regions have different landing depths.

16. The semiconductor structure according to claim 1, further comprising: a protection layer located between the stair step structure and the substrate.

17. The semiconductor structure according to claim 1, wherein an edge of the pad layer exceeds an edge of the conductive layer.

18. The semiconductor structure according to claim 1, wherein a horizontal spacing between two adjacent pad layers is substantially equal to a thickness of the pad layer.

19. A manufacturing method of a semiconductor structure, comprising: providing a substrate; and forming a pad structure on the substrate, wherein the pad structure comprises: material pairs stacked on the substrate to form a first stair step structure, wherein each of the material pairs comprises a conductive layer and a first dielectric layer located on the conductive layer; and pads, wherein each of the pads comprises: a conductive pillar embedded in the material pair and connected to the conductive layer of the material pair; and a pad layer located on the conductive pillar, wherein a thickness of the pad is greater than a thickness of the conductive layer.
